Received: 6 July 2022 | Accepted: 4 January 2023

DOI: 10.1111/biom.13824

BIOMETRIC METHODOLOGY

Biometrics WILEY

Optimal sampling for positive only electronic health record data

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Funding information
National Science Foundation, Grant/Award Number: DMS-1953527; National Institutes of Health, Grant/Award Numbers: R01-1RF1AG072272, R01-CA207365, R01-HG008980, R01-HL138306

Abstract

Identifying a patient's disease/health status from electronic medical records is a frequently encountered task in electronic health records (EHR) related research, and estimation of a classification model often requires a benchmark training data with patients' known phenotype statuses. However, assessing a patient's phenotype is costly and labor intensive, hence a proper selection of EHR records as a training set is desired. We propose a procedure to tailor the best training subsample with limited sample size for a classification model, minimizing its mean-squared phenotyping/classification error (MSE). Our approach incorporates "positive only" information, an approximation of the true disease status without false alarm, when it is available. In addition, our sampling procedure is applicable for training a chosen classification model which can be misspecified. We provide theoretical justification on its optimality in terms of MSE. The performance gain from our method is illustrated through simulation and a real-data example, and is found often satisfactory under criteria beyond MSE.

KEYWORDS

electronic health records, mean-squared error, optimal sampling, positive only

1 | INTRODUCTION

Identification of patients with the phenotype of interest is an essential step in utilizing electronic health records (EHR) data for research. Frequently, EHR are "positive-only", where we only have a small group of confirmed positive cases, while the rest of patients are left with unknown phenotype statuses. For example, patients with depression symptoms often need special treatment and management during their hospitalizations. At their admissions, physicians may use their EHR to assess their risk of depression, so they could choose proper interventions earlier to prevent deterioration and improve their chance of survival and recovery. The EHR of a patient diagnosed with depression may contain international classification of diseases (ICD) codes relevant to depression, which can help identify the patient as a case of having depression symptoms. However, patients whose EHRs do not contain depression-related ICD codes may still have the symptoms. In Gehrmann et al. (2018), a group of medical experts carefully reviewed clinical notes of 415 ICU patients from the MIMIC-III EHR data and identified 148 patients manifesting depression symptoms. Among the 148 patients, only 48 patients had depression diagnosis in their EHRs. This kind of positive only data, consisting of a small number of cases ("positives") and a large number of patients

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with unknown phenotypes ("unlabeled"), is common in EHR-based research (Zhang et al., 2020).

Because traditional manual chart review is labor-intensive hence infeasible for millions of patients' records in modern EHR data, it is common to rely on automatic algorithms to infer patients' phenotype statuses from their EHR (e.g., Hripcsak & Albers, 2013; Hou et al., 2020; Zhang et al., 2020). To develop and train such algorithms, gold standard phenotype statuses for a group of patients are needed. Due to the positive-only nature of EHR, the essential task here is to ascertain the phenotype statuses for a group of the unlabeled patients, and combine them with existing true-positive cases to form a training sample. Furthermore, the size of the training sample with complete labels will be substantially limited because phenotyping the unlabeled patients does require cost and labor intensive manual chart-review. These obstacles motivate the question of central interest to us—what is the optimal strategy for selecting the unlabeled patients? Here, the ultimate goal of the optimal sampling is to form a training dataset based on which the trained model achieves the optimal classification accuracy.

The subsampling technique has been widely used to relax computational burden or labor and experimental cost (Drineas et al., 2006). It aims to obtain a solution that is both efficient and practical by sampling a subset from a full dataset and performing estimation based on it. Recently, the optimal sampling design approaches have received more attention, which not only reduce the costs but further improve the efficiency of subsequent estimators. See Wang et al. (2018) and Wang and Ma (2021) for more detailed discussion. However, these procedures are not applicable in our context because the selection probabilities depend on both covariates and responses, while in our problem context, most of the responses are not available. On the other hand, Tan and Heagerty (2020) and Yin et al. (2022) proposed cost-effective procedures to handle health record data that minimize certain estimation criteria. However, these two papers focus on estimator augmentation after stratified sampling rather than optimal sampling design itself, and are specific to the logistic regression framework.

Unlike these existing literatures, we focus on devising a design that optimizes classification performance through rigorously investigating the sampling design based on the partially available information in combination with proper estimation procedure. Specifically, the key contributions of our work are the following. First, our proposed method determines optimal sampling weights by incorporating the surrogate indicator in the sampling scheme, when a given dataset has the positive-only nature. Second, our method is model-free in the sense that it is built on an arbitrary working classification model that does not have to be correctly specified. The method is flexible in that it can be applied to any working models. As a by-product, our method also yields the optimal model parameter estimation in terms of prediction MSE when the working model happens to be correct. In these perspectives, our method generalizes and improves the developments of Tan and Heagerty (2020) and Yin et al. (2022).

We now summarize the research question into a general statistical problem. Let $i = 1, \dots, N$. Let Y_i denote the latent binary phenotype of interest of patient i, such as a certain disease status. We denote $Y_i = 1$ if the patient has the disease, while $Y_i = 0$ if the patient does not have the disease. We equivalently name the diseased patient a case, and name the disease-free patient a control. Let $\mathbf{X}_i$ denote a vector of clinical variables that are predictive of Y_i and is fully observed. Let S_i be an observed binary variable. We name S_i an "anchor variable". S_i is linked to Y_i as the following. If $S_i = 1$, then $Y_i = 1$. If $S_i = 0$, then Y_i can be either 0 or 1. Note that this implies that there is no false alarm, that is, $\text{pr}(S_i = 1 \mid Y_i = 0) = 0$. We refer to the patients with $S_i = Y_i = 1$ as "anchor-positive cases", and those "$S_i = 0$" as "unlabeled" patients. In the motivating example, S_i is the binary indicator whether the patient has the diagnosis record of depression in EHR. Finally, we define R_i as an indicator of whether the ith patient will be selected into the training data and ascertain his or her disease status. By the definition of S_i, $S_i = 1$ implies that the ith patient has the disease, hence we automatically have $R_i = 1$ and $R_i Y_i = 1$. When $S_i = 0$, that is, if the ith observation is unlabeled and Y_i is missing, then R_i will be either 0 or 1 depending on whether the ith observation is selected into the training dataset for identifying its label Y_i. Accordingly, the observations with $R_i = 1$ will form a subsample with complete labels, and will hence be used to train the classification model. In summary, the observed data are $(\mathbf{X}_i, S_i, R_i, R_i Y_i), i = 1, \dots, N$. Without loss of generality, we assume $S_i = 0$ for $i = 1, \dots, N_0$ and $S_i = 1$ for $i = N_0 + 1, \dots, N$. Due to the nature of EHR, N is very large so we can view the data as the whole population, and view the quantities estimated based on the N observations as completely known.

Let a sampling probability be $\pi(\mathbf{X}_i) = \text{pr}(R_i = 1 \mid \mathbf{X}_i, S_i = 0)$. Our goal is to find a good $\pi(\cdot)$ so that we can randomly sample a subset according to $\pi(\mathbf{X}_i)$'s from the data $(\mathbf{X}_i, S_i = 0), i = 1, \dots, N_0$, obtain the corresponding Y_i's, combine with those with $S_i = 1$ hence $Y_i = 1$ to form a training dataset. Note that when $S_i = 1$, we do not need subsample since we know $Y_i = 1$. In a way, we can treat all the observations with $S_i = 1$ as sampled with probability 1 since they are included in the training data automatically. On the other hand, we do not have any Y_i values for those with $S_i = 0$ when perform sampling, so $\pi(\mathbf{X}_i)$ can only depend on $\mathbf{X}_i$ since this is all we have. To this end, we also define

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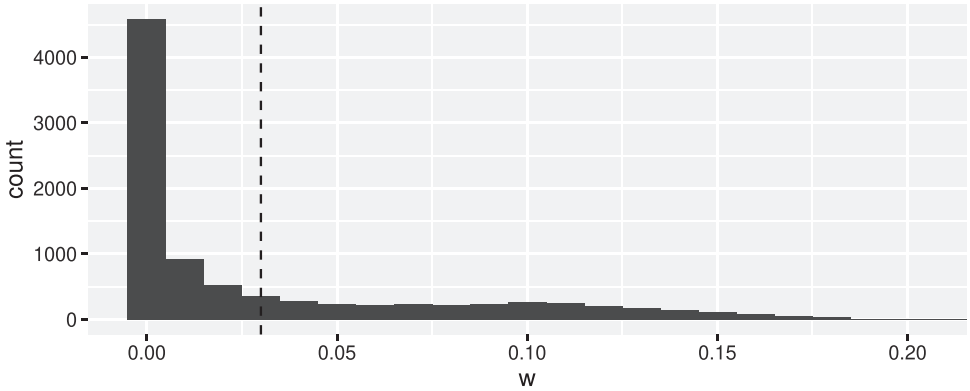

FIGURE 1 Histogram of sampling weights $w_i, i = 1, \dots, N_0$ in the simulation experiments with $N_0\pi_0 = 300$. Dotted line: sampling weight under simple random sampling

$\pi_T(\mathbf{X}_i, S_i) \equiv \text{pr}(R_i = 1 \mid \mathbf{X}_i, S_i) = (1 - S_i)\pi(\mathbf{X}_i) + S_i$, the probability of selecting any Y_i in the overall population (i.e., any patient in the EHR).

Recall that our eventual target is to perform classification on patients' phenotypes. For this purpose, let $p_0(\mathbf{X}) = \text{pr}(Y = 1 \mid \mathbf{X})$ and $\text{pr}(S = 1 \mid \mathbf{X}) = a(\mathbf{X})$. We can view $a(\mathbf{X})$ as known because it can be estimated based on the entire dataset with size N. Also, let $c(\mathbf{X}) \equiv \text{pr}(S = 1 \mid \mathbf{X}, Y = 1)$ be the anchor sensitivity. Note that $a(\mathbf{X}) = c(\mathbf{X})p_0(\mathbf{X})$. To perform classification, we will need a working model for $p_0(\mathbf{X})$, say $p(\mathbf{X}, \boldsymbol{\beta})$ and we will need to estimate $\boldsymbol{\beta}$, say the estimator is $\widehat{\boldsymbol{\beta}}$. Now, denote a new observation be $(\mathbf{X}^*, S^*, Y^*)$. We then classify Y^* using $p(\mathbf{X}^*, \widehat{\boldsymbol{\beta}})$ for a new observation if only $\mathbf{X}^*$ is available, and classify using $\text{pr}(Y^* = 1 \mid \mathbf{X}^*, S^* = 0) = \{p(\mathbf{X}^*, \widehat{\boldsymbol{\beta}}) - a(\mathbf{X}^*)\}/\{1 - a(\mathbf{X}^*)\}$ if in addition we have $S^* = 0$. In this paper, we propose an optimal sampling strategy $\pi(\mathbf{X}_i)$ that leads to a subsequent classification model with the highest classification accuracy.

2 | MAIN CONTRIBUTIONS

2.1 | Methodology

For simplicity, we first consider classifying the Y^* value of a future observation $\mathbf{X}^*$ without knowing the corresponding S^* value. Consider the case when $p(\mathbf{x}, \boldsymbol{\beta})$ is an approximate model in the whole EHR, that is, we allow $p_0(\mathbf{x}) \neq p(\mathbf{x}, \boldsymbol{\beta})$ for any $\boldsymbol{\beta}$. Our goal is to find a sampling strategy to minimize the mean-squared error (MSE) of $p(\mathbf{x}^*, \widehat{\boldsymbol{\beta}})$, that is, $E[\{Y^* - p(\mathbf{X}^*, \widehat{\boldsymbol{\beta}})\}^2]$. We can decompose $E[\{Y^* - p(\mathbf{X}^*, \widehat{\boldsymbol{\beta}})\}^2]$ into two parts, $E[\{Y^* - p_0(\mathbf{X}^*)\}^2]$ and $E[\{p_0(\mathbf{X}^*) - p(\mathbf{X}^*, \widehat{\boldsymbol{\beta}})\}^2]$. Since the first term is not relevant to the estimator $\widehat{\boldsymbol{\beta}}$, we aim at minimizing $E[\{p_0(\mathbf{X}^*) - p(\mathbf{X}^*, \widehat{\boldsymbol{\beta}})\}^2]$ which entails the discrepancy between the true model and the estimated model. We will show that this is an equivalent problem to minimizing

$$E\left\{\frac{p^2(\mathbf{X}, \boldsymbol{\beta}^*) + p_0(\mathbf{X}) - 2p_0(\mathbf{X})p(\mathbf{X}, \boldsymbol{\beta}^*)}{(1 - S)\pi(\mathbf{X}) + S}\mathbf{p}'_{\beta}{}^{\mathrm{T}}(\mathbf{X}, \boldsymbol{\beta}^*)\mathbf{A}^{-1}\mathbf{p}'_{\beta}(\mathbf{X}, \boldsymbol{\beta}^*)\right\} \tag{1}$$

with respect to a sampling strategy $\pi(\mathbf{X})$, where $\boldsymbol{\beta}^*$ is the unique minimizer of $E[\{p_0(\mathbf{X}) - p(\mathbf{X}, \boldsymbol{\beta})\}^2]$ with respect to $\boldsymbol{\beta}$, $\mathbf{A} \equiv E[\mathbf{p}'_{\beta}(\mathbf{X}, \boldsymbol{\beta}^*)^{\otimes 2} + \mathbf{p}''_{\beta\beta^{\mathrm{T}}}(\mathbf{X}, \boldsymbol{\beta}^*)\{p(\mathbf{X}, \boldsymbol{\beta}^*) - p_0(\mathbf{X})\}]$, and $\mathbf{p}'_{\beta}(\mathbf{x}, \boldsymbol{\beta})$ and $\mathbf{p}''_{\beta\beta^{\mathrm{T}}}(\mathbf{x}, \boldsymbol{\beta})$ are the gradient and the Hessian of $p(\mathbf{x}, \boldsymbol{\beta})$ with respect to $\boldsymbol{\beta}$.

Without any surprise, the optimal choice of $\pi(\mathbf{x})$ depends on the true $p_0(\mathbf{x})$ and $\boldsymbol{\beta}^*$, both are unknown. Hence, we need reasonable approximations of $p_0(\mathbf{x})$ and $\boldsymbol{\beta}^*$. Hereby, we assume that we are given a consistent estimator of $\boldsymbol{\beta}^*$, $\widetilde{\boldsymbol{\beta}}$. For example, the estimator $\widetilde{\boldsymbol{\beta}}$ can be obtained by minimizing the sample MSE of $p(\mathbf{x}, \boldsymbol{\beta})$, $m^{-1}\sum_{i=1}^{m}\{y_i - p(\mathbf{x}_i, \boldsymbol{\beta})\}^2$ with respect to $\boldsymbol{\beta}$ based on either external data or pilot study, $(\mathbf{x}_i, y_i), i = 1, \dots, m$.

Once we have $\widetilde{\boldsymbol{\beta}}$, the simplest thing to do is to replace $\boldsymbol{\beta}^*$ and $p_0(\mathbf{x})$ with $\widetilde{\boldsymbol{\beta}}$ and $p(\mathbf{x}, \widetilde{\boldsymbol{\beta}})$ respectively, and carry out the approximate optimization problem

$$\pi_{\text{opt}}(\mathbf{x}) = \text{argmin}_{\pi} E\left[\frac{p(\mathbf{X}, \widetilde{\boldsymbol{\beta}})\{1 - p(\mathbf{X}, \widetilde{\boldsymbol{\beta}})\}}{(1 - S)\pi(\mathbf{X}) + S}\mathbf{p}'_{\beta}{}^{\mathrm{T}}(\mathbf{X}, \widetilde{\boldsymbol{\beta}})\mathbf{A}^{-1}\mathbf{p}'_{\beta}(\mathbf{X}, \widetilde{\boldsymbol{\beta}})\right] \tag{2}$$

subject to the constraints $0 < \pi(\mathbf{x}) \leq 1$ and $E\{\pi(\mathbf{X})|S = 0\} \leq \pi_0$, where $\mathbf{A}$ is simplified to $E\{\mathbf{p}'_{\beta}(\mathbf{X}, \widetilde{\boldsymbol{\beta}})^{\otimes 2}\}$. Here, π_0 is a prefixed constant determined by the limit on resources. Because we consider the N observations as the whole population, writing $\mathbf{w} = (w_1, \dots, w_{N_0})^{\mathrm{T}}$, the above

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TABLE 1 Simulation results under correct model specification

$E\{p_0(\mathbf{X})\} = 0.05$							
Size	**Method**	**MSE**	**MSE$_0$**	**DEV**	**TN**	**TP**	**AUC**
	TRUE	0.025 (0.001)	0.021 (0.001)	1685.168	0.991	0.510	0.970
200	OPT	0.043 (0.009)	0.039 (0.009)	36826.695	0.975	0.549	0.947
	OSCA	0.067 (0.012)	0.066 (0.013)	4602.412	0.913	0.853	0.958
	SRS	0.050 (0.008)	0.046 (0.009)	58735.478	0.967	0.586	0.942
	MLE	0.059 (0.007)	0.059 (0.007)	4011.114	0.922	0.869	0.966
300	OPT	0.039 (0.008)	0.035 (0.007)	31249.481	0.979	0.539	0.951
	OSCA	0.051 (0.007)	0.050 (0.008)	3363.587	0.936	0.821	0.963
	SRS	0.044 (0.006)	0.040 (0.007)	47680.739	0.974	0.567	0.947
	MLE	0.049 (0.005)	0.048 (0.005)	3267.891	0.937	0.840	0.967
400	OPT	0.036 (0.006)	0.032 (0.006)	23624.540	0.981	0.542	0.955
	OSCA	0.043 (0.005)	0.041 (0.006)	2809.119	0.948	0.793	0.965
	SRS	0.040 (0.006)	0.036 (0.006)	38676.695	0.978	0.549	0.951
	MLE	0.044 (0.004)	0.042 (0.004)	2870.093	0.946	0.817	0.967
$E\{p_0(\mathbf{X})\} = 0.20$							
Size	**Method**	**MSE**	**MSE$_0$**	**DEV**	**TN**	**TP**	**AUC**
	TRUE	0.063 (0.002)	0.063 (0.002)	4094.214	0.957	0.732	0.958
200	OPT	0.084 (0.015)	0.086 (0.015)	24863.828	0.942	0.721	0.945
	OSCA	0.131 (0.030)	0.140 (0.027)	9129.010	0.805	0.898	0.930
	SRS	0.098 (0.014)	0.100 (0.014)	51564.708	0.935	0.720	0.940
	MLE	0.103 (0.007)	0.109 (0.008)	6830.639	0.846	0.917	0.954
300	OPT	0.075 (0.010)	0.077 (0.011)	12224.434	0.946	0.725	0.949
	OSCA	0.104 (0.011)	0.110 (0.012)	6775.164	0.849	0.889	0.945
	SRS	0.081 (0.012)	0.083 (0.013)	20707.036	0.945	0.721	0.947
	MLE	0.091 (0.005)	0.096 (0.006)	5959.228	0.869	0.899	0.955
400	OPT	0.070 (0.006)	0.072 (0.007)	6414.891	0.950	0.725	0.951
	OSCA	0.090 (0.006)	0.094 (0.007)	5776.053	0.874	0.876	0.950
	SRS	0.072 (0.007)	0.074 (0.008)	7771.140	0.949	0.724	0.951
	MLE	0.084 (0.004)	0.088 (0.005)	5491.735	0.883	0.885	0.956

Note: Mean and standard deviation (in parentheses) of the corresponding summaries.

reduces to

$$\mathbf{w}_{\text{opt}} = \text{argmin}_{\mathbf{w}} \sum_{i=1}^{N_0} \frac{p(\mathbf{x}_i, \widetilde{\boldsymbol{\beta}})\{1 - p(\mathbf{x}_i, \widetilde{\boldsymbol{\beta}})\}}{w_i} \mathbf{p}'_{\boldsymbol{\beta}}{}^{\mathrm{T}} (\mathbf{x}_i, \widetilde{\boldsymbol{\beta}}) \mathbf{A}_N^{-1} \mathbf{p}'_{\boldsymbol{\beta}}(\mathbf{x}_i, \widetilde{\boldsymbol{\beta}}) \tag{3}$$

subject to the constraints $0 < w_i \leq 1$ for $i = 1, \ldots, N_0$ and $\sum_{i=1}^{N_0} w_i \leq N_0 \pi_0$, where $\mathbf{A}_N \equiv N^{-1} \sum_{i=1}^{N} \mathbf{p}'_{\boldsymbol{\beta}}(\mathbf{x}_i, \widetilde{\boldsymbol{\beta}})^{\otimes 2}$. Note that in Equation (3), the minimizer clearly is obtained at $E\{\pi(\mathbf{X})|S = 0\} = \pi_0$, that is, $\sum_{i=1}^{N_0} w_i = N_0 \pi_0$. So, we can restrict the search region on the boundary.

Once we have $\mathbf{w}_{\text{opt}}$, we sample R_i from Bernoulli(w_i) for $i = 1, \ldots, N_0$ and set $R_i = 1$ for $i = N_0 + 1, \ldots, N$ in order to obtain the observations $(\mathbf{X}_i, S_i, R_i, R_i Y_i), i = 1, \ldots, N$. In other words, we measure the Y_i values of the observations with $(S_i, R_i) = (0, 1)$. Then, we solve the estimating equation

$$\sum_{i=1}^{N} \frac{R_i}{(1 - S_i) w_i + S_i} \mathbf{p}'_{\boldsymbol{\beta}}(\mathbf{x}_i, \boldsymbol{\beta}) \{p(\mathbf{x}_i, \boldsymbol{\beta}) - Y_i\} = \mathbf{0} \tag{4}$$

with respect to $\boldsymbol{\beta}$.

Note that given any fixed sampling weights $w_i, i = 1, \ldots, N_0$, solving Equation (4), that is, minimizing the inverse probability weighted sample MSE, provides the most efficient estimator for $\boldsymbol{\beta}^*$ under the situation $p(\mathbf{x}, \boldsymbol{\beta}^*) = p_0(\mathbf{x})$ (Qin et al., 2017). The most efficient estimator $\widehat{\boldsymbol{\beta}}$ will also lead to the minimum MSE of $p(\mathbf{x}, \widehat{\boldsymbol{\beta}})$. Indeed, theoretical analysis in Section 2.2 will show that our proposed procedure guarantees the optimal classification MSE among all possible estimators of $\boldsymbol{\beta}$. Note that an equivalent representation of the sample MSE in the binary classification context is $N^{-1} \sum_{i=1}^{N} R_i / \{(1 - S_i) w_i +$

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TABLE 2 Simulation results under model misspecification

$E\{p_0(\mathbf{X})\} = 0.05$							
Size	**Method**	**MSE**	**MSE_0**	**DEV**	**TN**	**TP**	**AUC**
	TRUE	0.023 (0.001)	0.020 (0.001)	1575.514	0.991	0.548	0.975
200	OPT	0.046 (0.009)	0.041 (0.009)	47652.551	0.974	0.525	0.937
	OSCA	0.073 (0.011)	0.073 (0.011)	4930.425	0.903	0.858	0.956
	SRS	0.051 (0.009)	0.046 (0.009)	60229.612	0.968	0.552	0.932
	MLE	0.067 (0.008)	0.067 (0.008)	4483.136	0.911	0.864	0.960
300	OPT	0.042 (0.006)	0.037 (0.006)	43313.776	0.978	0.511	0.941
	OSCA	0.056 (0.007)	0.055 (0.008)	3682.277	0.928	0.820	0.959
	SRS	0.046 (0.007)	0.041 (0.007)	53151.853	0.974	0.530	0.938
	MLE	0.055 (0.005)	0.055 (0.006)	3640.138	0.928	0.831	0.961
400	OPT	0.039 (0.006)	0.035 (0.006)	35424.364	0.980	0.502	0.944
	OSCA	0.048 (0.006)	0.046 (0.006)	3109.903	0.941	0.787	0.961
	SRS	0.043 (0.006)	0.038 (0.006)	45619.777	0.978	0.515	0.941
	MLE	0.049 (0.004)	0.048 (0.005)	3204.222	0.939	0.805	0.962
$E\{p_0(\mathbf{X})\} = 0.20$							
Size	**Method**	**MSE**	**MSE_0**	**DEV**	**TN**	**TP**	**AUC**
	TRUE	0.058 (.002)	0.059 (0.002)	3750.032	0.959	0.761	0.965
200	OPT	0.089 (.015)	0.093 (0.013)	17583.095	0.938	0.685	0.933
	OSCA	0.142 (.014)	0.153 (0.014)	9778.256	0.784	0.914	0.932
	SRS	0.105 (.017)	0.107 (0.016)	44193.354	0.930	0.687	0.928
	MLE	0.119 (.008)	0.128 (0.008)	7994.974	0.819	0.916	0.945
300	OPT	0.080 (.008)	0.085 (0.007)	7823.384	0.943	0.690	0.939
	OSCA	0.115 (.008)	0.124 (0.009)	7542.984	0.827	0.897	0.940
	SRS	0.086 (.012)	0.090 (0.011)	14564.369	0.940	0.689	0.936
	MLE	0.106 (.006)	0.113 (0.006)	6935.105	0.844	0.896	0.946
400	OPT	0.077 (.005)	0.082 (0.005)	5519.307	0.946	0.690	0.941
	OSCA	0.101 (.006)	0.108 (0.007)	6506.819	0.852	0.879	0.943
	SRS	0.078 (.006)	0.083 (0.006)	6339.479	0.944	0.691	0.940
	MLE	0.098 (.005)	0.104 (0.005)	6357.604	0.861	0.880	0.946

Note: Mean and standard deviation (in parentheses) of the corresponding summaries.

TABLE 3 Simulation results under violation of the positive-only assumption

Size	**Method**	**MSE**	**MSE_0**	**DEV**	**TN**	**TP**	**AUC**
	TRUE	0.063 (0.002)	0.065 (0.002)	4094.214	0.957	0.732	0.958
200	OPT	0.082 (0.015)	0.086 (0.016)	19821.625	0.938	0.734	0.945
	OSCA	0.355 (0.017)	0.374 (0.019)	18803.531	0.202	1.000	0.923
	SRS	0.095 (0.016)	0.099 (0.016)	41650.395	0.932	0.729	0.940
	MLE	0.365 (0.016)	0.385 (0.017)	20007.290	0.269	1.000	0.942
300	OPT	0.073 (0.009)	0.076 (0.009)	8205.711	0.944	0.736	0.950
	OSCA	0.269 (0.014)	0.280 (0.016)	14712.217	0.402	0.998	0.939
	SRS	0.078 (0.012)	0.081 (0.012)	15439.427	0.942	0.733	0.948
	MLE	0.288 (0.013)	0.301 (0.015)	15885.078	0.408	0.998	0.947
400	OPT	0.069 (0.005)	0.072 (0.005)	5172.247	0.947	0.738	0.952
	OSCA	0.218 (0.012)	0.225 (0.013)	12387.180	0.538	0.993	0.946
	SRS	0.071 (0.006)	0.073 (0.007)	6453.707	0.946	0.736	0.951
	MLE	0.239 (0.011)	0.248 (0.012)	13411.293	0.510	0.995	0.950

Note: Mean and standard deviation (in parentheses) of the corresponding summaries.

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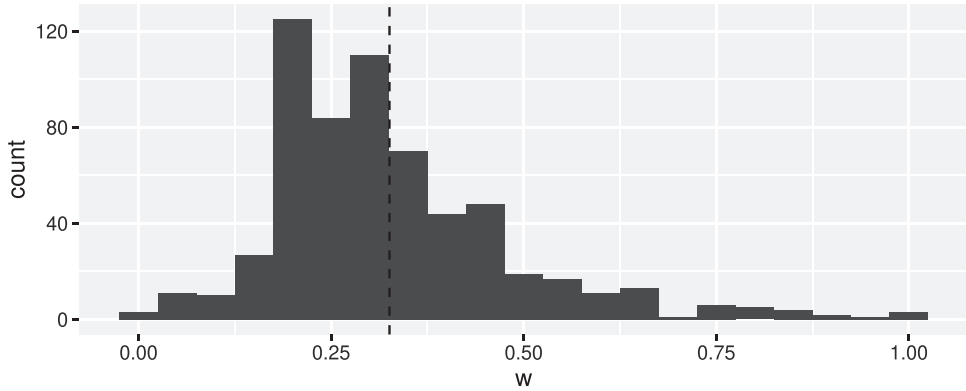

FIGURE 2 Histogram of sampling weights $w_i, i = 1, \dots, N_0$ in the data example with $m = 100$ and $N_0\pi_0 = 200$. Dotted line: sampling weight under simple random sampling

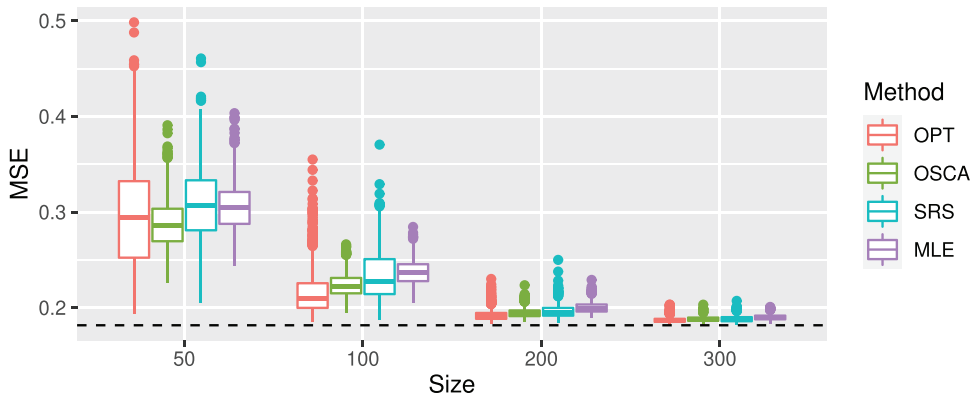

FIGURE 3 Boxplots of mean-squared error (MSE) in the data example with $m = 100$. Dashed: MSE based on the full dataset

$S_i\}[[Y_i\{1 - p(\mathbf{x}_i, \boldsymbol{\beta})\}^2 + (1 - Y_i)p(\mathbf{x}_i, \boldsymbol{\beta})^2]$, hence we can interpret minimizing MSE as minimizing the total squared mis-classification probabilities. In addition, our estimation procedure can also be viewed as a likelihood-based approach where a normal working model $Y \sim N\{p(\mathbf{x}, \boldsymbol{\beta}), \sigma^2\}$ is adopted. Although the response variable is binary, we can adopt a working model for a general response because our procedure does not require the model to be correct. Empirical evidences will be demonstrated through simulation studies and a data example in Sections 3 and 4 to support the theory. Even if $p(\mathbf{x}, \boldsymbol{\beta}^*) \neq p_0(\mathbf{x})$, using $p(\mathbf{x}, \boldsymbol{\beta}^*)$ as a working model for $p_0(\mathbf{x})$ is the most sensible choice, which leads to Equation (4) as well, and it still yields a consistent estimator for $\boldsymbol{\beta}^*$.

We now write out the algorithm explicitly.

(1) Form $\mathbf{A}_N = N^{-1} \sum_{i=1}^{N} \mathbf{p}'_{\boldsymbol{\beta}}(\mathbf{x}_i, \widetilde{\boldsymbol{\beta}})^{\otimes 2}$ and obtain the sampling weights as

$$\widehat{\mathbf{w}}_{\text{opt}} = \text{argmin}_{\mathbf{w}} \sum_{i=1}^{N_0} \frac{p(\mathbf{x}_i, \widetilde{\boldsymbol{\beta}})\{1 - p(\mathbf{x}_i, \widetilde{\boldsymbol{\beta}})\}}{w_i} \mathbf{p}'_{\boldsymbol{\beta}}{}^{\mathrm{T}} (\mathbf{x}_i, \widetilde{\boldsymbol{\beta}}) \mathbf{A}_N^{-1} \mathbf{p}'_{\boldsymbol{\beta}}(\mathbf{x}_i, \widetilde{\boldsymbol{\beta}}) \tag{5}$$

subject to the constraints $0 < w_i \leq 1$ for $i = 1, \dots, N_0$ and $\sum_{i=1}^{N_0} w_i = N_0\pi_0$.

(2) Sample R_i from Bernoulli$(\widehat{w}_i)$ for $i = 1, \dots, N_0$ and set $R_i = 1$ for $i = N_0 + 1, \dots, N$. Then, determine the value of Y_i through extensive manual chart reviews if $(S_i, R_i) = (0, 1)$.

(3) Obtain $\widehat{\boldsymbol{\beta}}$ as

$$\widehat{\boldsymbol{\beta}} = \text{argmin}_{\boldsymbol{\beta}} \sum_{i=1}^{N} \frac{R_i}{(1 - S_i)\widehat{w}_i + S_i} \{y_i - p(\mathbf{x}_i, \boldsymbol{\beta})\}^2. \tag{6}$$

(4) Perform classification of Y^* at a new observation $\mathbf{X}^*$ as $\widehat{Y}^* = I\{p(\mathbf{X}^*, \widehat{\boldsymbol{\beta}}) > 0.5\}$.

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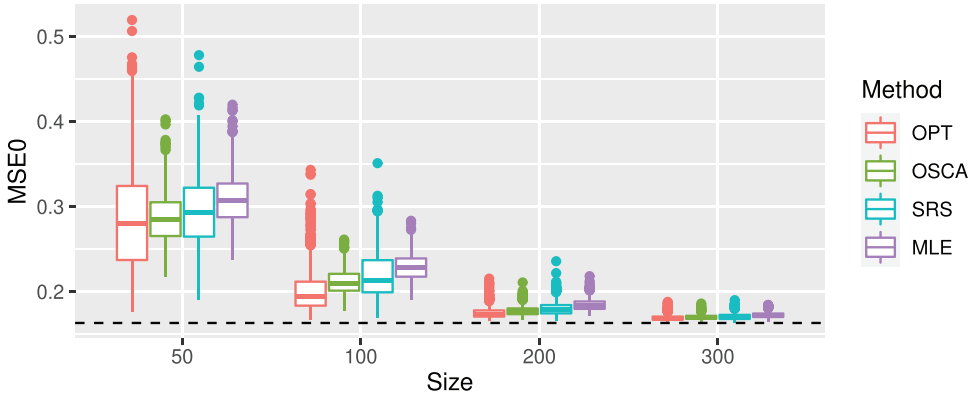

FIGURE 4 Boxplots of MSE_0 in the data example with $m = 100$. Dashed: MSE_0 based on the full dataset

Remark 1. We point out that under the constraint $\sum_{i=1}^{N_0} w_i = N_0\pi_0$ alone, the solution $\mathbf{w}_{\text{opt}}$ in Equation (3) is

$$w_i = N_0\pi_0 \frac{\sqrt{c_i}}{\sum_{i=1}^{N_0}\sqrt{c_i}} \tag{7}$$

for $i = 1, \dots, N_0$, where $c_i \equiv p(\mathbf{x}_i, \widetilde{\boldsymbol{\beta}})\{1 - p(\mathbf{x}_i, \widetilde{\boldsymbol{\beta}})\}\mathbf{p}'_{\beta}{}^{\mathrm{T}}(\mathbf{x}_i, \widetilde{\boldsymbol{\beta}})\mathbf{A}_N^{-1}\mathbf{p}'_{\beta}(\mathbf{x}_i, \widetilde{\boldsymbol{\beta}})$. Considering the additional constraint $0 < w_i \le 1$, we note that Equation (3) has a closed form solution (7) if $0 < N_0\pi_0\sqrt{c_i}/\sum_{i=1}^{N_0}\sqrt{c_i} \le 1$ for all $i = 1, \dots, N_0$.

We end this section with some notes on practical implementation of our method. First, the cutoff value of 0.5 in Step 4 of the algorithm can be modified to any other appropriate value. We may take into account additional information such as the commonness/rareness of disease in the population to adjust the cutoff to be larger/smaller than 0.5. For example, we can set the cutoff value to be the average disease rate in the sample, that is, $N^{-1}\sum_{i=1}^{N} R_i Y_i/\{(1 - S_i)w_i + S_i\}$ (Agresti, 2003). Furthermore, the cutoff value may be guided by the intended application of the classification model. For instance, if the purpose of the study is for identifying high-risk individuals, the choice can be guided by the desired level of accuracy. In addition, if the pilot data are available to us, it is beneficial to include it. To this end, we can form a pooled dataset combining the pilot data and the data we have been considering so far. We then conduct the similar procedure on the combined data, while assigning $\pi_T(\mathbf{X}_i, S_i) = 1$ and $R_i = 1$ for each observation in the pilot data.

2.2 | Theory

Now, we study the theoretical properties of the estimation procedure stated above. All proofs are relegated to Web Appendices. We require the following regularity conditions:

1. $\boldsymbol{\beta}^*$ is the unique minimizer of $E[\{p_0(\mathbf{X}) - p(\mathbf{X}, \boldsymbol{\beta})\}^2]$.
2. $\mathbf{A} \equiv E[\mathbf{p}'_{\beta}(\mathbf{X}^*, \boldsymbol{\beta}^*)^{\otimes 2} + \{p(\mathbf{X}^*, \boldsymbol{\beta}^*) - p_0(\mathbf{X}^*)\}\partial^2 p(\mathbf{X}^*, \boldsymbol{\beta}^*)/\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^{\mathrm{T}}]$ is nonsingular.

Condition 1 is very mild and it implies that $\boldsymbol{\beta}^*$ is a solution to $E[\mathbf{p}'_{\beta}(\mathbf{X}, \boldsymbol{\beta})\{p_0(\mathbf{X}) - p(\mathbf{X}, \boldsymbol{\beta})\}] = \mathbf{0}$. It often holds naturally under appropriate parameterization when the model $p(\mathbf{x}, \boldsymbol{\beta})$ is true. Condition 2 naturally holds when the model $p(\mathbf{X}, \boldsymbol{\beta}^*)$ is not drastically misspecified, and is a reasonable requirement in both theory and practice.

From the description in Section 2.1, it is clear that the estimator of $\boldsymbol{\beta}$ relies on the sampling weights. Hence, in the following, when we emphasize this dependence, we write $\widehat{\boldsymbol{\beta}}_{\mathbf{w}}$, and reserve $\widehat{\boldsymbol{\beta}}$ for $\widehat{\boldsymbol{\beta}}_{\mathbf{w}_{\text{opt}}}$.

Theorem 1. *Let $\widehat{\boldsymbol{\beta}}_{\mathbf{w}}$ be the solution of Equation (4) and $\mathbf{w}_{\text{opt}}$ be as defined in Equation (3). Then under any given $\mathbf{w}$, $\widehat{\boldsymbol{\beta}}_{\mathbf{w}}$ minimizes the estimated classification MSE. Furthermore, when $p(\mathbf{x}, \boldsymbol{\beta}^*) = p_0(\mathbf{x})$ and $\widetilde{\boldsymbol{\beta}} = \boldsymbol{\beta}^*$, $\mathbf{w}_{\text{opt}}$ minimizes the MSE of $p(\mathbf{X}^*, \widehat{\boldsymbol{\beta}}_{\mathbf{w}})$, that is, $E[\{Y^* - p(\mathbf{X}^*, \widehat{\boldsymbol{\beta}}_{\mathbf{w}_{\text{opt}}})\}^2] \le E[\{Y^* - p(\mathbf{X}^*, \widehat{\boldsymbol{\beta}}_{\mathbf{w}})\}^2]$ for any $\mathbf{w}$ satisfying $0 < w_i \le 1$ for $i = 1, \dots, N_0$ and $\sum_{i=1}^{N_0} w_i \le N\pi_0$.*

Theorem 2. *Assume the regularity conditions specified in Robins et al. (1994) and let $\widehat{\boldsymbol{\beta}}_{\mathbf{w}}$ be the solution of Equation (4). When $p(\mathbf{x}, \boldsymbol{\beta}^*) = p_0(\mathbf{x})$, $E[\{Y^* - p(\mathbf{X}^*, \widehat{\boldsymbol{\beta}}_{\mathbf{w}})\}^2] \le E[\{Y^* - p(\mathbf{X}^*, \widehat{\boldsymbol{\beta}}_{\mathbf{w}})\}^2]$ for any estimator $\widehat{\boldsymbol{\beta}}_{\mathbf{w}}$ and $\mathbf{w}$ satisfying $0 < w_i \le 1$ for $i = 1, \dots, N_0$ and $\sum_{i=1}^{N_0} w_i \le N\pi_0$.*

Theorem 1 states that our optimal sampling choice $\mathbf{w}_{\text{opt}}$ yields the estimator $\widehat{\boldsymbol{\beta}}_{\mathbf{w}_{\text{opt}}}$ with the minimum MSE among the class of estimators $\widehat{\boldsymbol{\beta}}_{\mathbf{w}}$ from Equation (4) associated with an arbitrary sampling weight $\mathbf{w}$. On the other hand,

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Theorem 2 states that given a sampling strategy $\mathbf{w}$, an estimator $\widehat{\boldsymbol{\beta}}_{\mathbf{w}}$ obtained by solving Equation (4) has a smaller classification MSE compared to any other estimator $\widehat{\boldsymbol{\beta}}_{\mathbf{w}}$. Therefore, we conclude that $\widehat{\boldsymbol{\beta}}_{\mathbf{w}_{\text{opt}}}$ provides the minimum MSE among all possible estimators $\widehat{\boldsymbol{\beta}}_{\mathbf{w}}$. We highlight this conclusion as a remark below.

Remark 2. Theorem 1 illustrates the optimality of the sampling weights $\mathbf{w}_{\text{opt}}$ in terms of minimizing the classification MSE when the working model is true. Obviously, in practice, a working model almost always serves as an approximate model, hence the optimality is also only approximate. This is inevitable as we always need to pay a price for what we are unable to know. On the other hand, the MSE also relies on the choice of the model paramater estimator. To this end, as established in Theorem 2, our estimator $\widehat{\boldsymbol{\beta}}$ is also optimal in the sense that it minimizes the MSE to the leading order, and is efficient when the working model is correct, and is consistent when the working model is misspecified. The optimality of $\widehat{\boldsymbol{\beta}}_{\mathbf{w}}$ translates to the local optimality of the MSE under any weight choice $\mathbf{w}$, and to the global optimality of the MSE under $\mathbf{w}_{\text{opt}}$. We summarize this result as Corollary 1.

Corollary 1. *Let $\mathbf{w}_{\text{opt}}$ be as defined in Equation (3) and $\widehat{\boldsymbol{\beta}}_{\mathbf{w}_{\text{opt}}}$ be the solution of Equation (4). Among all the choices of sampling weights $\mathbf{w}$ and all the consistent estimators of $\boldsymbol{\beta}^*$ that does not require the working model $p(\mathbf{x}, \boldsymbol{\beta})$ to be correct, $\widehat{\boldsymbol{\beta}}_{\mathbf{w}_{\text{opt}}}$ minimizes the classification MSE if $p(\mathbf{x}, \boldsymbol{\beta}^*) = p_0(\mathbf{x})$ and $\widetilde{\boldsymbol{\beta}} = \boldsymbol{\beta}^*$.*

We finally also point out the role that the external information based estimator $\widetilde{\boldsymbol{\beta}}$ plays in Corollary 2. Essentially, as long as $\widetilde{\boldsymbol{\beta}}$ is a consistent estimator, it is as good as the true $\boldsymbol{\beta}$.

Corollary 2. *Let $\mathbf{w}_{\text{opt}}$ be as defined in Equation (3) and $\widehat{\boldsymbol{\beta}}_{\mathbf{w}_{\text{opt}}}$ be the solution of Equation (4). Given that $\widetilde{\boldsymbol{\beta}}$ converges to $\boldsymbol{\beta}^*$ in probability, $\widehat{\boldsymbol{\beta}}_{\mathbf{w}_{\text{opt}}}$ minimizes the classification MSE among all choices of $\mathbf{w}$ and $\widehat{\boldsymbol{\beta}}_{\mathbf{w}}$ if $p(\mathbf{x}, \boldsymbol{\beta}^*) = p_0(\mathbf{x})$.*

Since the optimal sampling weights $\mathbf{w}_{\text{opt}}$ aims at minimizing Equation (3) which includes $\widetilde{\boldsymbol{\beta}}$, it is obvious that $\mathbf{w}_{\text{opt}}$ depends on the estimate $\widetilde{\boldsymbol{\beta}}$ from external data or pilot study. However, Corollary 2 points out that even when $\widetilde{\boldsymbol{\beta}} \neq \boldsymbol{\beta}^*$, our choice of $\mathbf{w}_{\text{opt}}$ and the corresponding $\widehat{\boldsymbol{\beta}}_{\mathbf{w}_{\text{opt}}}$ still results in the minimum classification MSE as long as $\widetilde{\boldsymbol{\beta}}$ is consistent for $\boldsymbol{\beta}^*$.

2.3 | Classification of Y^* given $S^* = 0$

We now extend the classification based on $\mathbf{X}^*$ alone to the case when the anchor variable is known to be 0. By conditioning on S, we can decompose $p_0(\mathbf{x}) = \text{pr}(Y = 1|\mathbf{x})$ as $\text{pr}(Y = 1|\mathbf{x}, S = 1)\text{pr}(S = 1|\mathbf{x}) + \text{pr}(Y = 1|\mathbf{x}, S = 0)\text{pr}(S = 0|\mathbf{x})$, where $\text{pr}(Y = 1|\mathbf{x}, S = 1) = 1$ and $\text{pr}(S = 1|\mathbf{x}) = a(\mathbf{x})$. In addition, as we have pointed out, $a(\mathbf{x})$ is known. Hence, $\text{pr}(Y = 1|\mathbf{x}, S = 0)$ can be estimated using $p(\mathbf{x}, \widehat{\boldsymbol{\beta}})$ by

$$\widehat{\text{pr}}(Y = 1|\mathbf{x}, S = 0) = \frac{p(\mathbf{x}, \widehat{\boldsymbol{\beta}}) - a(\mathbf{x})}{1 - a(\mathbf{x})}. \tag{8}$$

Then, perform classification of Y^* at a new observation $(\mathbf{X}^*, S^* = 0)$ as $\widehat{\mathbf{Y}}^* = I\{\widehat{\text{pr}}(Y^* = 1|\mathbf{X}^*, S^* = 0) > 0.5\}$. In other words, we follow the same procedure in the algorithm described in Section 2.1, while replace the probability in the sixth step with Equation (8).

3 | SIMULATION EXPERIMENTS

We conduct simulation studies to assess the finite sample performance of our method. For comparison, we implemented our proposed optimal sampling in combination with optimal estimator of $\boldsymbol{\beta}$ (OPT), the estimation procedure proposed by Yin et al. (2022) (OSCA), the simple random sampling in combination with solving Equation (6) (SRS), and the maximum likelihood estimator of the model $p(\mathbf{x}, \boldsymbol{\beta})$ under simple random sampling (MLE). Note that MLE mimics a naive method not considering the missingness probability, and SRS mimics the usual inverse probability weighting approach, both under the simple random sampling over the observations with $S = 0$. Summary statistics are also calculated under the true model (TRUE) to serve as a benchmark.

All results below are based on 1,000 replicates with sample size $N = 10,000$ for each replicate. In addition, we considered different constraints by setting $N_0\pi_0 = 200, 300$, and 400 to further compare the performances under different limits of resources. For the entire replicates, we fixed $\widetilde{\boldsymbol{\beta}}$ estimated based on 400 observations drawn from the same population and independent of the N observations. Lastly, we set the average disease rate $E\{p_0(\mathbf{X})\} = 0.05$ and 0.2, and let the anchor sensitivity to be about 40%, that is, $\text{pr}(S = 1 \mid Y = 1) = 0.4$. The detail of the simulation design in each experiment is individually described below.

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We report the MSE of the full dataset, calculated as $N^{-1}\sum_{i=1}^{N}\{y_i - p(\mathbf{x}_i, \widehat{\boldsymbol{\beta}})\}^2$, and the MSE in the subset of the data with $S = 0$ (MSE_0), calculated as $N_0^{-1}\sum_{i=1}^{N_0}\{y_i - \text{pr}(Y_i = 1 \mid \mathbf{x}_i, S_i = 0, \widehat{\boldsymbol{\beta}})\}^2$, where $\text{pr}(Y = 1|\mathbf{x}, S = 0)$ is estimated using Equation (8) or is set to zero if Equation (8) leads to a negative value. Although our estimator is designed to optimize MSE, we also report deviance (DEV), the full data specificity (TN), and sensitivity (TP), where the classification is $I\{p(\mathbf{x}, \widehat{\boldsymbol{\beta}}) > 0.5\}$, and the area under the receiver operator characteristic (ROC) curve (AUC). The deviance is calculated based on N_ϵ observations , whose predicted probabilities are within $[10^{-10}, 1 - 10^{-10}]$ and multiplied by N/N_ϵ.

3.1 | Correct model specification

We first consider the situation when the classification model is correctly specified. The covariate $\mathbf{X}$ is of dimension 12 with X_1, X_5, X_9 generated independently from $N(0, 1)$, X_2, X_6, X_{10} from discrete uniform on $\{1, 2, 3, 4, 5\}$, X_3, X_7, X_{11} from Bernoulli with a success rate $p = 0.5$, and X_4, X_8, X_{12} from Bernoulli with $p = 0.1$ being categorical variables with low prevalence. We set the regression coefficient $\boldsymbol{\beta} = (0.2, 0.3, 0.4, -0.5, 0.8, 1.0, -1.2, 1.4, 1.7, -2.0, 2.3, 2.6)^{\mathrm{T}}$, so that X_1, X_2, X_3, X_4 represent weak, X_5, X_6, X_7, X_8 moderate, and $X_9, X_{10}, X_{11}, X_{12}$ strong covariates.

Further, we draw Y_i from a logistic regression model with $p_0(\mathbf{x}_i) = p(\mathbf{x}_i, \boldsymbol{\beta}) = \text{expit}(\beta_0 + \boldsymbol{\beta}^{\mathrm{T}}\mathbf{x}_i)$, where $\text{expit}(\eta) \equiv 1/(1 + \mathrm{e}^{-\eta})$ and β_0 is chosen to yield the average disease rates 5% and 20%, respectively. Finally, the anchor variable S_i is generated as the product of Y_i and a random variable distributed as Bernoulli with $p = \text{expit}(\gamma_0 + \boldsymbol{\gamma}^{\mathrm{T}}\mathbf{x}_i)/\text{expit}(\beta_0 + \boldsymbol{\beta}^{\mathrm{T}}\mathbf{x}_i)$, where $\boldsymbol{\gamma} = \boldsymbol{\beta} + 0.1\text{sign}(\boldsymbol{\beta})$ and γ_0 is chosen to ensure that approximately 40% of the patients are detected by the anchor variable S. This implies that S given only $\mathbf{x}$ follows a Bernoulli distribution with $p = \text{expit}(\gamma_0 + \boldsymbol{\gamma}^{\mathrm{T}}\mathbf{x})$, thus we use $a(\mathbf{x}) = \text{expit}(\gamma_0 + \boldsymbol{\gamma}^{\mathrm{T}}\mathbf{x})$ to perform classification as described in Section 2.3.

To illustrate the estimation procedure of OPT, in Figure 1, we first plot the histogram of the sampling weights estimated by solving Equation (3) in one simulation. As can be seen, OPT leads to very different sampling weights for different observations. This contrasts with the simple sampling weight, which is fixed as π_0. As described in Theorem 1 and Corollary 2, this sampling strategy will result in forming validation data that leads to a subsequent estimator that is optimal in MSE, which will be illustrated next.

In Table 1, we summarize how the implemented methods perform under various simulation settings. We can observe that OPT indeed shows the best performance in terms of both MSE and MSE_0. Compared to SRS, the OPT weight choice leads to improved performance of $\widehat{\boldsymbol{\beta}}_{\mathbf{w}}$, in line with Theorem 1. The optimal performance of OPT is more clear to see when $N_0\pi_0 = 200$, which suggests that the improvement of OPT over the estimator $\widehat{\boldsymbol{\beta}}_{\mathbf{w}}$ is more evident if the available resource is more restricted. We can also see that OPT provides the most similar result to TRUE in terms of both TN and TP among all the methods implemented, even though we do not expect OPT to show any superior performance in terms of sensitivity and specificity. In contrast, OSCA and MLE perform the worst in terms of MSE and MSE_0 even though the model is correctly specified, while they are better in DEV and AUC.

3.2 | Wrong model specification

To further assess the performance where the classification model is misspecified, we generate the covariates $\mathbf{X}^* = (\mathbf{X}^{\mathrm{T}}, X_1^2, X_2X_6, X_7X_{11}, X_8X_{12})^{\mathrm{T}}$, where $\mathbf{X}$ is drawn from the same generation mechanism described in Section 3.1. In addition, we set the regression coefficient $\boldsymbol{\beta}^*$ to be a concatenation of $\boldsymbol{\beta}$ given in Section 3.1 and additional coefficients corresponding to higher order terms, that is, $\boldsymbol{\beta}^* = (\boldsymbol{\beta}^{\mathrm{T}}, -0.4, -0.6, 0.8, 1.0)^{\mathrm{T}}$. Then, we generate Y_i from a Bernoulli distribution with the success rate $p_0(\mathbf{x}_i) = \text{expit}(\beta_0 + \boldsymbol{\beta}^{*\mathrm{T}}\mathbf{x}_i^*)$ where β_0 is again chosen to achieve the disease rate 5% and 20%. To reflect the misspecified classification model scenario, the model $p(\mathbf{x}, \boldsymbol{\beta})$ is fitted without the higher order covariates, that is, it only contains $\mathbf{x}$. Finally, the anchor variable S_i is generated as a multiplication of Y_i and a random variable distributed as Bernoulli with $\text{expit}(\gamma_0 + \boldsymbol{\gamma}^{*\mathrm{T}}\mathbf{x}_i^*)/\text{expit}(\beta_0 + \boldsymbol{\beta}^{*\mathrm{T}}\mathbf{x}_i^*)$, so that $S|\mathbf{x}^*$ follows Bernoulli with $p = \text{expit}(\gamma_0 + \boldsymbol{\gamma}^{*\mathrm{T}}\mathbf{x}^*)$. Here, $\boldsymbol{\gamma}^* = \boldsymbol{\beta}^* + 0.1\text{sign}(\boldsymbol{\beta}^*)$ and γ_0 is chosen to ensure that 40% of patients are detected by the anchor variable S. In this setting, we use $a(\mathbf{x}^*) = \text{expit}(\gamma_0 + \boldsymbol{\gamma}^{*\mathrm{T}}\mathbf{x}^*)$ for classification as proposed in Section 2.3.

In Table 2, we summarize the performances of the implemented methods. Even though the model is misspecified hence the assumption of Corollary 2 is violated, the result still shows that our method OPT estimates the model with the smallest MSE and MSE_0, thus provides the best possible classification among all the methods. Compared to SRS, the MSE of $\widehat{\boldsymbol{\beta}}_{\mathbf{w}}$ is better for OPT, which possibly suggests that the optimal sampling weight choice is robust to the model specification. Again, we can see that when $N_0\pi_0$ is smaller, the gain of OPT over the other meth-

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TABLE 4 Data example results

$m = 100$							
Size	**Method**	**MSE**	**MSE_0**	**DEV**	**TN**	**TP**	**AUC**
	FULL	0.182 (0.000)	0.163 (0.000)	739.200	0.926	0.300	0.714
50	OPT	0.295 (0.056)	0.284 (0.061)	3511.233	0.772	0.431	0.613
	OSCA	0.288 (0.026)	0.287 (0.030)	1521.553	0.520	0.656	0.624
	SRS	0.309 (0.039)	0.297 (0.042)	4369.568	0.775	0.435	0.609
	MLE	0.306 (0.025)	0.309 (0.029)	1629.172	0.475	0.686	0.625
100	OPT	0.217 (0.024)	0.202 (0.026)	1173.383	0.860	0.385	0.661
	OSCA	0.224 (0.013)	0.212 (0.015)	1069.759	0.763	0.500	0.659
	SRS	0.234 (0.026)	0.219 (0.026)	1658.405	0.841	0.410	0.651
	MLE	0.238 (0.014)	0.229 (0.016)	1128.305	0.704	0.554	0.656
200	OPT	0.192 (0.006)	0.175 (0.007)	802.642	0.907	0.350	0.688
	OSCA	0.195 (0.005)	0.178 (0.006)	826.873	0.887	0.374	0.686
	SRS	0.196 (0.007)	0.180 (0.008)	860.564	0.896	0.368	0.686
	MLE	0.200 (0.006)	0.185 (0.007)	847.391	0.847	0.441	0.682
300	OPT	0.187 (0.003)	0.169 (0.003)	768.319	0.921	0.330	0.696
	OSCA	0.188 (0.003)	0.170 (0.003)	768.789	0.919	0.319	0.698
	SRS	0.188 (0.003)	0.171 (0.004)	783.098	0.914	0.346	0.699
	MLE	0.190 (0.003)	0.172 (0.003)	776.052	0.899	0.379	0.695
$m = 200$							
Size	**Method**	**MSE**	**MSE_0**	**DEV**	**TN**	**TP**	**AUC**
	FULL	0.170 (0.000)	0.153 (0.000)	590.416	0.929	0.361	0.743
50	OPT	0.282 (0.054)	0.271 (0.058)	3140.056	0.788	0.450	0.635
	OSCA	0.262 (0.026)	0.261 (0.030)	1186.123	0.618	0.637	0.665
	SRS	0.291 (0.036)	0.280 (0.039)	3691.941	0.792	0.454	0.632
	MLE	0.276 (0.025)	0.279 (0.029)	1272.175	0.576	0.678	0.669
100	OPT	0.204 (0.025)	0.190 (0.027)	980.448	0.872	0.411	0.688
	OSCA	0.204 (0.012)	0.192 (0.015)	823.034	0.814	0.492	0.696
	SRS	0.222 (0.024)	0.209 (0.025)	1460.681	0.854	0.441	0.676
	MLE	0.215 (0.013)	0.206 (0.015)	867.045	0.765	0.558	0.698
200	OPT	0.178 (0.005)	0.162 (0.006)	634.448	0.911	0.390	0.723
	OSCA	0.181 (0.004)	0.164 (0.005)	656.411	0.903	0.386	0.720
	SRS	0.184 (0.007)	0.169 (0.008)	704.734	0.903	0.406	0.713
	MLE	0.184 (0.004)	0.168 (0.005)	668.724	0.873	0.455	0.722
300	OPT	0.173 (0.002)	0.156 (0.002)	604.041	0.922	0.377	0.734
	OSCA	0.175 (0.002)	0.157 (0.002)	613.060	0.923	0.347	0.733
	SRS	0.176 (0.002)	0.159 (0.003)	623.377	0.918	0.392	0.729
	MLE	0.176 (0.002)	0.158 (0.002)	616.538	0.907	0.398	0.733

Note: Mean and standard deviation (in parentheses) of the corresponding summaries.

ods is larger in terms of MSE. In addition, TN and TP results suggest that OPT has a similar level of specificity to the true model, though sensitivity is not as good, possibly caused by the wrong model choice. Lastly, as in Section 3.1, MSE and MSE_0 of OSCA and MLE are worse than that of OPT, while their DEV and AUC are competitive. Interestingly, OPT achieves the best AUC when the sampling size is 400.

3.3 | Robustness to violation of positive-only assumption

To assess the sensitivity of our proposal to the positive-only assumption, we now consider the case, where true data structure does not satisfy $P(Y = 1 \mid S = 1) = 1$. The covariates $\mathbf{X}_i$ and the response Y_i are drawn in the same manner as in Section 3.1. Then, the anchor variable S_i is defined

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as $Y_i S_{1i} + (1 - Y_i) S_{0i}$, where S_{1i} follows the Bernoulli distribution with $p_i = \text{expit}(\gamma_0 + \boldsymbol{\gamma}^{\mathrm{T}} \mathbf{x}_i)/\text{expit}(\beta_0 + \boldsymbol{\beta}^{\mathrm{T}} \mathbf{x}_i)$ and S_{0i} is drawn from Bernoulli with probability $q = 0.025$. Here, β_0 is chosen to yield the disease rate 20%, and $\boldsymbol{\gamma} = \boldsymbol{\beta} + 0.1\text{sign}(\boldsymbol{\beta})$, where γ_0 is chosen to satisfy $P(S = 1|Y = 1) = 0.4$. This data generation procedure leads to 20% false alarm, that is, $P(Y = 1 \mid S = 1) = 0.8$.

We implement all methods under the false assumption $P(Y = 1 \mid S = 1) = 1$. That is, we mistakenly label all the Y's to be 1 when the corresponding anchor variable $S = 1$. We then implement all estimation procedures using the logistic regression model of Y given $\mathbf{X}$, and measure the prediction performance using the true Y.

Table 3 summarizes the performances of implemented methods under violation of the positive only assumption. The results show that OPT has the best performance among the implemented methods in both MSE and MSE_0. Compared to the results in Table 1, both OPT and SRS are robust to the mislabeling, while the performances of two likelihood-based approaches, OSCA and MLE, severely deteriorated due to the mislabeling in terms of MSE and MSE_0. We find that this is due to the weight difference between observations corresponding to $S = 1$ and $S = 0$. In OPT and SRS, the weights assigned to observations with $S = 1$ are much larger than those assigned to observations with $S = 0$, hence the mislabeled observations, which must have $S = 1$, have almost no impact on the final estimation. Interestingly, OPT also achieves the best DEV and AUC when the sampling size is above 300, even though its goal is to minimize the MSE.

4 | DATA EXAMPLE

To further illustrate the numerical performance of our method, we apply it to the motivating example, where we try to identify patients with depression symptoms at their hospital admissions using their EHRs. The data were part of the Medical Information Mart for Intensive Care III database version 1.4 (MIMIC-III v1.4) (Johnson et al., 2016), a freely-available database on health-related information of over 40,000 patients who stayed in critical care units of the Beth Israel Deaconess Medical Center between 2001 and 2012. We used a subset of MIMIC-III that consists of 767 admission entries from 415 frequent ICU patients. The same set of patients and admission entries has been analyzed in Gehrmann et al. (2018). We emphasize that our method in practice can be conducted on bigger databases, as demonstrated in Section 3 with larger sample size. Sampling weights in Equation (3) can be obtained by off-the-shelf algorithms for constrained optimization, and have a closed form under suitable conditions as given in Remark 1.

Our goal is to classify admission entries into two groups, whether patients present any depression symptoms during hospital admissions. Hence, we defined a binary label $Y = 1$ if a patient had any depression symptoms. In addition, for the anchor variable S, we extracted the ICD-9 diagnosis codes to determine whether a patient was diagnosed as having depression. In addition, a panel of medical experts in Gehrmann et al. (2018) has conducted cross-validated chart reviews of those entries, and identified 30.7% of the entries presenting depression symptoms (i.e., $Y = 1$). Among these patients with $Y = 1$, 24.4% were diagnosed as having depression (i.e., $S = 1$), hence anchor-positive. There were 5 observations with $S = 1$ but $Y = 0$, which violates the positive-only assumption. These were deleted from the dataset in our analysis. We then used additional information from the records as covariates, which includes gender, age, ethnicity, marital status, insurance type, admission type, previous length of stays in hospital, number of procedures done on patients during the admission in each category (evaluation and management, medicine, radiology, surgery), and the number of prescribed medications for depressions (Amitriptyline, Clomipramine, Desipramine, Doxepin, Duloxetine, Escitalopram, Fluoxetine, Imipramine, Levomilnacipran, Nortriptyline, Paroxetine, Sertraline, Venlafaxine).

Using the above information, we implemented our method (OPT) as if the labels Y were not observed. For comparison, we also implemented the procedure proposed by Yin et al. (2022) (OSCA), the simple random sampling in combination with solving Equation (6) (SRS), and the maximum likelihood estimator under the simple random sampling (MLE). As a benchmark, the minimizer of MSE using the full dataset (FULL) was also implemented. For all methods, we adopted the logistic regression model with all the covariates mentioned above and age squared. No additional higher order terms or interaction terms of the covariates are included. For our method, we used $m = 100$ and 200 observations sampled from the full dataset to obtain the initial estimate $\widetilde{\boldsymbol{\beta}}$. We then implemented all the methods on the remaining data.

To compare the performance of the methods, we report the prediction MSE and the MSE on the observations with $S = 0$ (MSE_0). Although our method is designed to minimize MSE, to assess the performance of our method under other criteria, deviance (DEV), the specificity (TN), the sensitivity (TP), and the AUC were also reported. The criteria are calculated based on the data excluding the m observations used for initial estimation. Details on how the measurements were calculated can be found in Section 3, and the anchor positive probability $a(\mathbf{x})$ was fitted by the logistic regression on S given the same covariates. The summaries are based on 1,000 replicates, and we consider different sampling sizes $N_0\pi_0 = 50, 100, 200, 300$.

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Figure 2 illustrates the sampling weights estimated by our method when $m = 100$ and $N_0\pi_0 = 200$. The figure shows that our procedure yields various sampling weights for different observations, in contrast to the simple random sampling, which would assign the same sampling weight 0.326 to every observation. This weighting strategy indeed improved MSE and MSE_0 of the following estimates, as can be seen from Figures 3 and 4. These two figures show the boxplots of MSE and MSE_0 under each allowed subsampling size. We can see that our method indeed results in estimators with smaller MSE and MSE_0 than other methods in general, regardless of the subsampling sizes.

For more accurate comparison, we summarized the average and the standard deviation of each criterion in Table 4. From the table, we can see that OPT indeed has the smallest MSE and MSE_0 among all methods under reasonably large subsample sizes. Indeed, when the subsample size is very small such as $N_0\pi_0 = 50$, every method performs worse than a pure random guess, which yields MSE 0.25. Also, it can be seen from Table 4 that under smaller subsample size, the gain from our method is more prominent. This suggests that our method is especially valuable when resource is very limited. On the other hand, in terms of specificity and sensitivity, OPT provides the most similar results to the benchmark results by FULL in all situations. In terms of DEV and AUC, although OPT performs less satisfactory when the sampling size is very small (50), it is very similar to the best performance when the size increases to 100, and almost has the best performance when it further increases to 200 and 300. Lastly, the pilot size m has almost no impact on the competitiveness of OPT, suggesting that a consistent $\widetilde{\boldsymbol{\beta}}$ is sufficient as shown in Corollary 2.

5 | DISCUSSION

We have proposed an optimal subsampling procedure in terms of prediction MSE. The procedure does not require a correct prediction model and can be conducted when the allowed subsampling size is restricted. Although the detailed derivation aims to minimize MSE, the idea and general approach are applicable to any other criteria. Under different criterions, the optimal sampling weights are also likely different. Theoretical derivations are also criterion dependent and may be challenging especially when the chosen criterion is non-smooth or discontinuous. Our derivation is conducted under a single prediction model. If several models are of interest, we recommend to construct an ensemble model which includes these models as special cases to increase model flexibility and carry out the proposed procedure.

We also point out that our method can be directly applied to the simpler case, where no anchor variable S is available. To do so, we simply treat the no anchor variable case as a special case with an anchor variable, but all anchor variables are zero, that is, $S_i = 0$ for all $i = 1, \ldots, N$, and $N_0 = N$. Then, our method is immediately applicable. Because our method is applicable regardless whether there is a genuine positive-only anchor variable, in the case when we are given an anchor variable that is not necessarily positive-only, we recommend dismissing it or treating it as a covariate to remain in our framework, even though the method appears to be robust in our numerical experiments. Throughout the paper, we have considered a binary response Y. However, our analysis procedure also can be applied to the case when the response is continuous. On the other hand, in the case of a binary response, criteria other than MSE are also of interest, such as misclassification rate. We believe that each criterion requires a specifically tailored procedure and poses its own challenge in establishing the optimality.

ACKNOWLEDGMENTS

The work is partially supported by NSF grant DMS-1953527 and NIH grants R01-HL138306, R01-CA207365, R01-1RF1AG072272, R01-HG008980.

DATA AVAILABILITY STATEMENT

The data that support the findings in this paper are available in the Supporting information of this paper.

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SUPPORTING INFORMATION

Web Appendices referenced in Sections 2.2, and R codes implementing the simulation experiments in Section 3 and the data example in Section 4 are available with this paper at the Biometrics website on Wiley Online Library.

How to cite this article: Lee, S.-H., Ma, Y., Wei, Y. & Chen, J. (2023) Optimal sampling for positive only electronic health record data. *Biometrics*, 79, 2974–2986. https://doi.org/10.1111/biom.13824

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